INCH-POUND

MIL-DTL-83503/23C
w/AMENDMENT 1
4 August 2010
SUPERSEDING
MIL-DTL-83503/23C
21 July 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL PRINTED WIRING BOARD SOLDER TRANSITION FOR TERMINATING ROUND CONDUCTORS ON .050 INCH CENTERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83503.

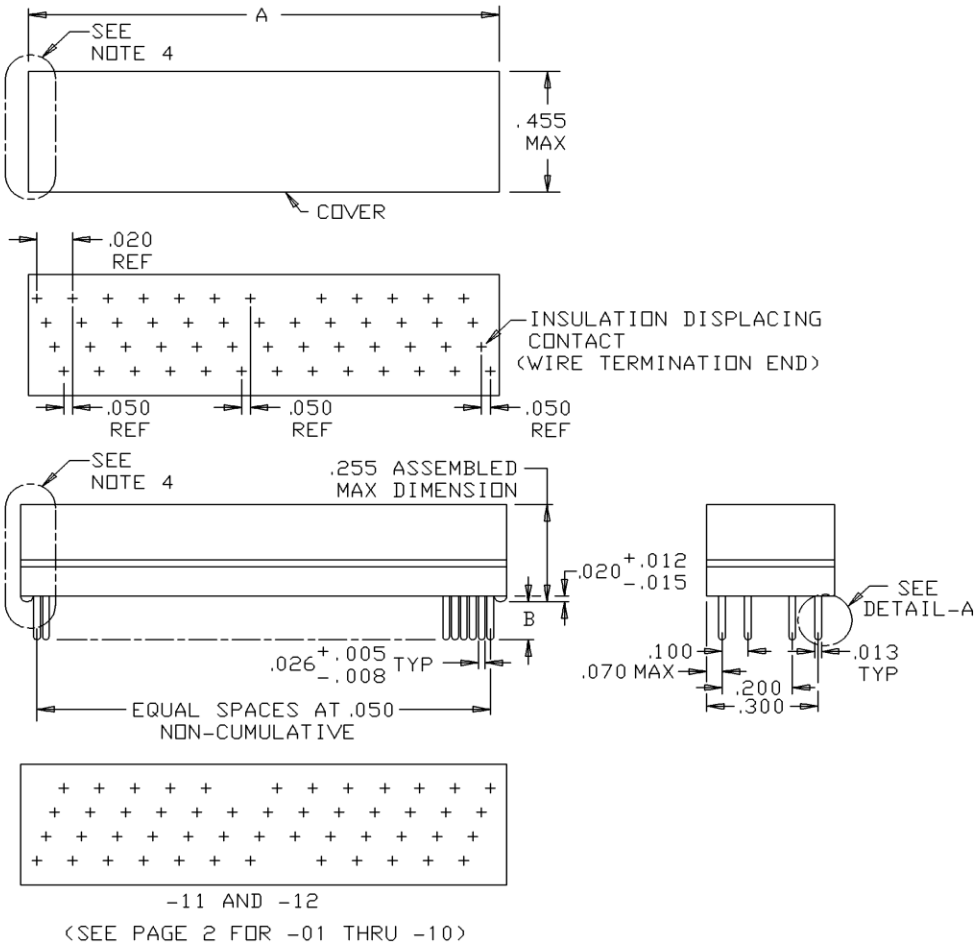


FIGURE 1. Dimensions and configuration.

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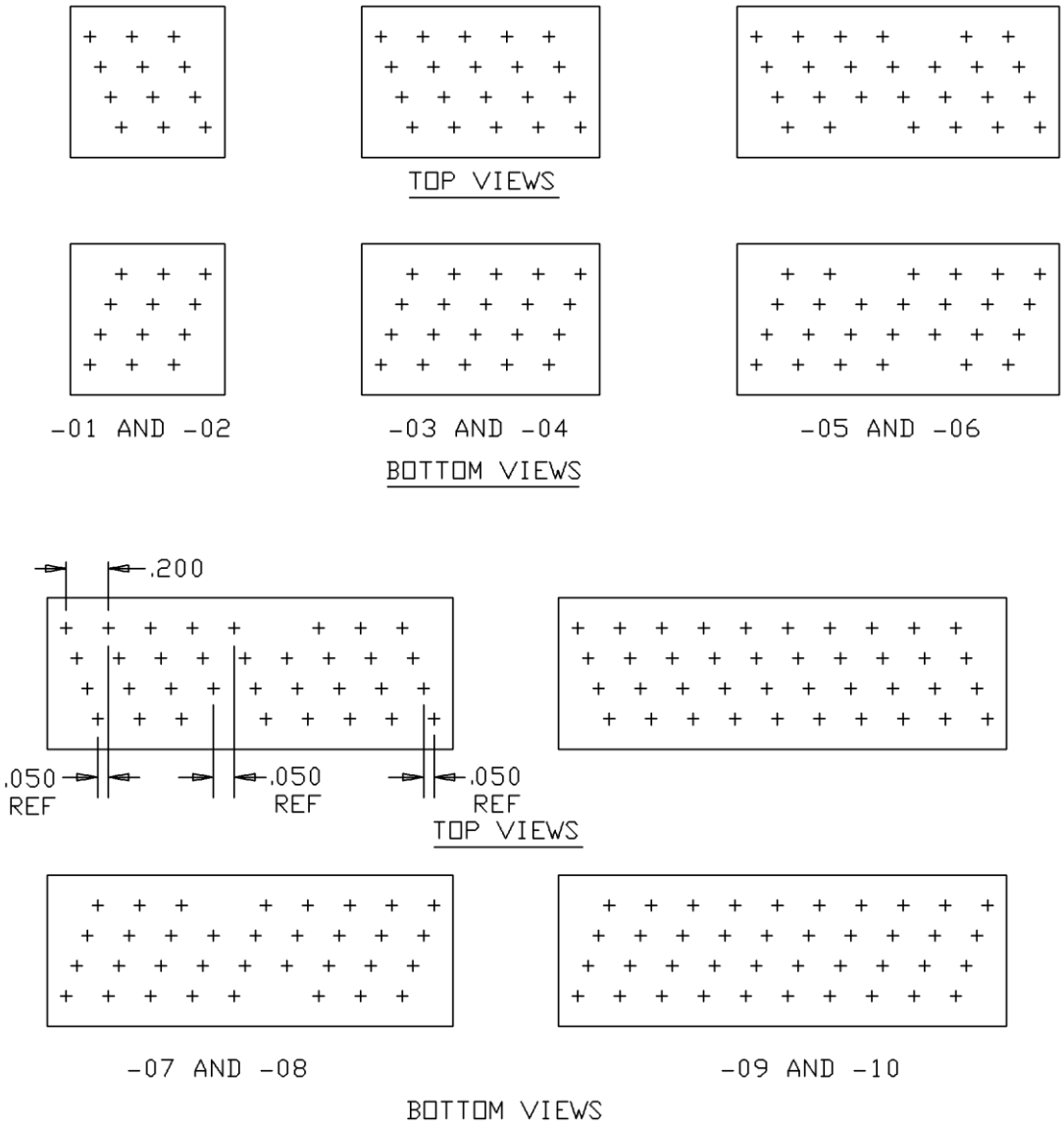
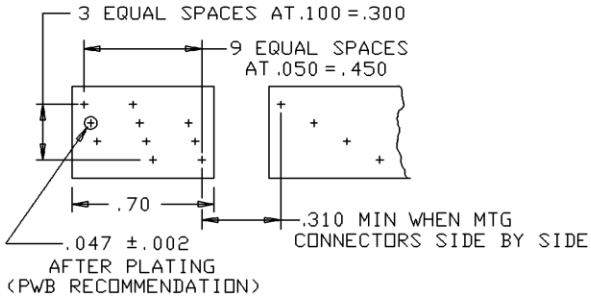
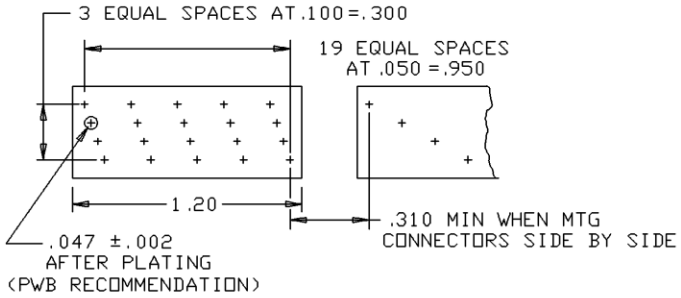


FIGURE 1. Dimensions and configuration - Continued.

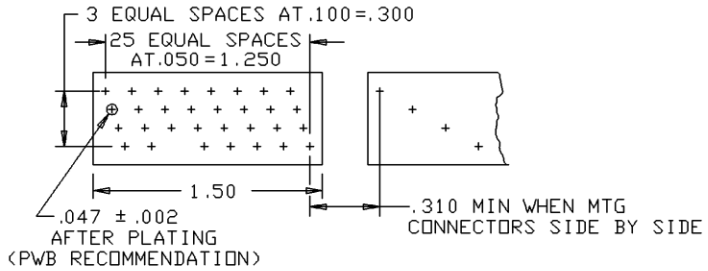
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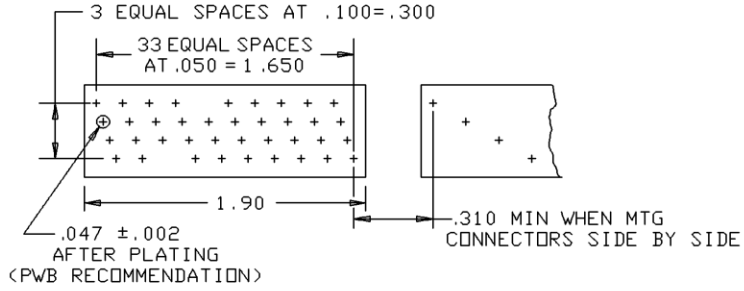
M83503/23 -01 AND -02



M83503/23 -03 AND -04



M83503/23 -05 AND -06



M83503/23 -07 AND -08

FIGURE 1. Dimensions and configuration - Continued.

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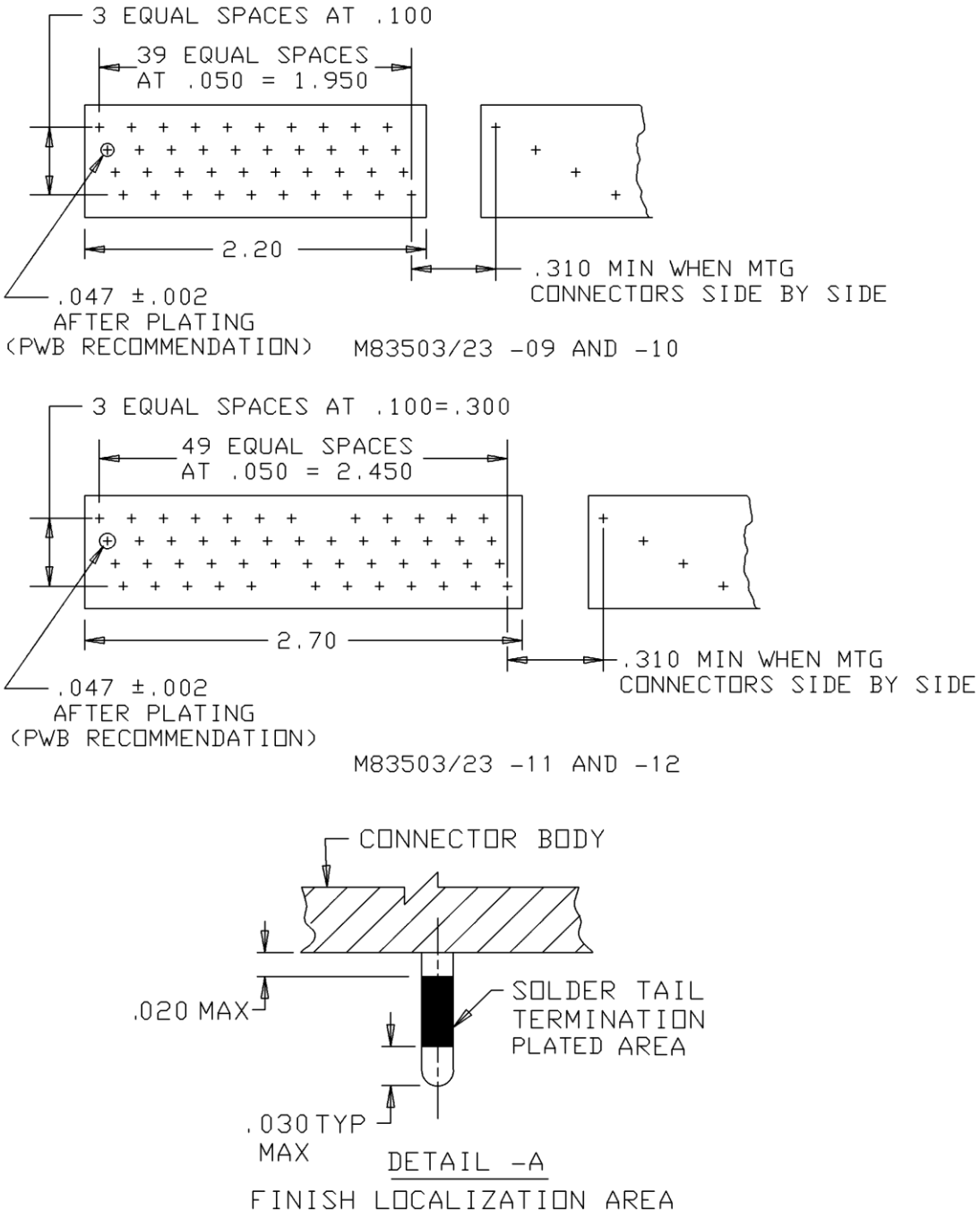


FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.047	1.19	.950	24.13
.005	0.13	.050	1.27	1.20	30.5
.008	0.20	.070	1.78	1.250	31.75
.012	0.30	.100	2.54	1.50	38.1
.013	0.33	.200	5.08	1.650	41.91
.015	0.38	.255	6.48	1.90	48.3
.020	0.51	.300	7.62	1.950	49.53
.026	0.66	.310	7.87	2.20	55.9
.030	0.76	.450	11.43	2.450	62.23
		.455	11.56	2.70	68.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm) for three place decimals and ± .02 inch (0.5 mm) for two place decimals.
4. The method used to connect cover to body is optional, but shall meet the requirements of MIL-DTL-83503 and this specification sheet.

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number	Number of positions	Dimension	
		A inches (mm) max	B inches (mm) max
-01	10	.72 (18.3)	.100 (2.54)
-02	10	.72 (18.3)	.156 (3.96)
-03	20	1.22 (31.0)	.100 (2.54)
-04	20	1.22 (31.0)	.156 (3.96)
-05	26	1.52 (38.6)	.100 (2.54)
-06	26	1.52 (38.6)	.156 (3.96)
-07	34	1.92 (48.8)	.100 (2.54)
-08	34	1.92 (48.8)	.156 (3.96)
-09	40	2.22 (56.4)	.100 (2.54)
-10	40	2.22 (56.4)	.156 (3.96)
-11	50	2.72 (69.1)	.100 (2.54)
-12	50	2.72 (69.1)	.156 (3.96)

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1 and table I.

Temperature range: -55° to +120°C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Contacts shall be phosphor bronze or beryllium copper. Phosphor bronze shall be in accordance with ASTM-B139/B139M spring temper or brass in accordance with CDA 260 spring temper. Beryllium copper shall be in accordance with ASTM-B196/B196M or ASTM-B197/B197M alloy UNS No. C17200 spring temper.

Plating: Tin-lead in the termination area in accordance with MIL-DTL-83503.

Housing: Glass field polyester in accordance with MIL-M-24519, type GPT-15F, GPT-20F, GPT-30F.

Wire accommodation: 28 AWG stranded on .050 (1.27 mm) inch spacing between conductor centerlines.

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Voltage rating: 300 volts.

Test wire size: 28 AWG stranded.

Preconditioning: Not applicable.

Withstanding voltage:

Sea level: 500 V rms, minimum.

Altitude: 200 V rms, minimum.

Cable retention: Not applicable.

Contact resistance: 50 milliohms, maximum.

Mating and unmating force: Not applicable.

Durability: Not applicable.

Interface:

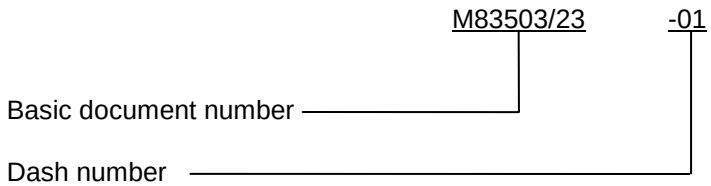
Round conductor flat cable: .050 inch (1.27 mm) centers to .062 inch (1.57 mm).

Printed wiring boards (thickness): .093 inch (2.36 mm) to .125 inch (3.18 mm).

Marking:

Part or Identifying Number (PIN): The PIN shall consist of the letter “M”, followed by the basic specification number, “/”, the specification sheet number, “-” and the dash number in table I.

Example:



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Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-DTL-83503, this document references the following:

MIL-M-24519
ASTM-B139/B139M
ASTM-B196/B196M
ASTM-B197/B197M
CDA 260

CONCLUDING MATERIAL

Custodians:
Air Force - 85
DLA - CC

Review activity:
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-2010-137)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil

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